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Application of quantum Monte Carlo integration to Markovian backward stochastic differential equations

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Abstract
In this paper, we introduce a novel Quantum Least-Squares Monte Carlo (QLSM) algorithm for solving backward stochastic differential equations (BSDEs). The QLSM algorithm leverages quantum computing to enhance the efficiency of traditional least-squares Monte Carlo methods. We present the detailed procedure of the QLSM method and an evaluation of its accuracy and computational cost. This approach has significant potential for applications in mathematical finance, particularly in pricing complex financial derivatives and risk management.

Keywords Markovian BSDE, least-square Monte Carlo, quantum algorithm

Research Activity Group Mathematical Finance

1. Preliminaries

We begin by introducing some basic notation and Markovian backward stochastic differential equations (BSDEs). In what follows, we fix $T \in (0, \infty)$, the terminal time, and a filtered probability space $(\Omega, \mathscr{F}, P, \mathbb{F})$, $\mathbb{F} := \{\mathscr{F}_t\}_{t\in[0,T]}$, equipped with a d-dimensional standard $\mathscr{F}_t$-Brownian motion $\{W_t\}_{t\in[0,T]}$ starting from 0, where $\mathscr{F}_t := \sigma(W_u; u \in [0, t])$ $(t \in [0, T])$.

1.1 Notation

For $A \in \mathbb{R}^{n\times d}$, $\sigma_{\min}(A)$ denotes the minimal singular value of A. For $x \in \mathbb{R}$, $[x]$ denotes the set of all natural numbers less than or equal to x, and $[x]_{+0} := [x] \cup \{0\}$. $\mathrm{Var}(X)$ denotes the variance of a random variable X. For $t \in [0, T]$ and $p \in [1, \infty)$, $L^p_t(\mathbb{R}^{n\times d})$ denotes the set of $\mathbb{R}^{n\times d}$-valued $\mathcal{F}_t$-measurable random variables X such that $\|X\|_{L^p} := (\mathbb{E}[|X|^p])^{1/p} < \infty$. Furthermore, $L^{p,q}_{\mathbb{F}}(\mathbb{R}^{n\times d})$ denotes the space of $\mathbb{F}$-adapted $\mathbb{R}^{n\times d}$-valued stochastic processes $\phi := (\phi_t)_{t\in[0,T]}$ such that $\|\phi\|_{L^{p,q}_{\mathbb{F}}(\mathbb{R}^{n\times d})} := \mathbb{E}\left[\left(\int_0^T |\phi(t)|^q dt\right)^{p/q}\right]^{1/p} < \infty$, and $H^p_{\mathbb{F}}(\mathbb{R}^{n\times d})$ denotes the space of continuous $\mathbb{F}$-adapted $\mathbb{R}^{n\times d}$-valued stochastic processes $\phi := (\phi_t)_{t\in[0,T]}$ such that $\|\phi\|_{H^p_{\mathbb{F}}(\mathbb{R}^{n\times d})} := \mathbb{E}\Big[\sup_{t\in[0,T]} |\phi(t)|^p\Big]^{1/p} < \infty$.

1.2 Markovian backward stochastic differential equations

Let b: $[0, T] \times \mathbb{R}^{d_1} \to \mathbb{R}^{d_1}$, σ: $[0, T] \times \mathbb{R}^{d_1} \to \mathbb{R}^{d_1\times d}$, f: $[0, T] \times \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \times \mathbb{R}^{d_2\times d} \to \mathbb{R}^{d_2}$, and g: $\mathbb{R}^{d_1} \to \mathbb{R}^{d_2}$ be given. Consider the following system of stochastic differential equations,

$$\begin{cases} dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t, & X_0 = x, \\ dY_t = -f(t, X_t, Y_t, Z_t)dt + Z_t dW_t, & Y_T = g(X_T), \end{cases} \tag{1}$$

for $t \in [0, T]$. We call (1) a Markovian BSDE. Note that the first SDE in (1) for X is solved independently, forward in time. Using its solution $(X_t)_{t\in[0,T]}$, the second equation in (1) is solved backward in time. Now, let us introduce the basic result of the solution of the Markovian BSDE (1).

Assumption 1 (i) *b, σ, f, and g are uniformly Lipschitz continuous in (x, y, z) with a Lipschitz constant L.*
(ii) $\int_0^T |b(t, 0)|dt + \int_0^T |\sigma(t, 0)|^2 dt + \int_0^T |f(t, 0, 0, 0)|dt < \infty$.
(iii) *b, σ, and f are uniformly Hölder - 1/2 continuous in t with a Hölder constant L.*

Theorem 2 ([1, Theorems 3.3.1, 3.4.3, 4.3.1, and 4.4.4]) *Let Assumption 1 (i) and (ii) hold. Then, for some $p \geq 2$, Markovian BSDE (1) has a unique adapted L^p-solution $(X, Y, Z) \in H^p_{\mathbb{F}}(\mathbb{R}^{d_1}) \times H^p_{\mathbb{F}}(\mathbb{R}^{d_2}) \times L^{p,2}_{\mathbb{F}}(\mathbb{R}^{d_2\times d})$.*

2. Classical numerical method

In this section, we introduce the best known numerical method for solving BSDEs, the least-square regression (LSR). Due to the computational complexity involved, we will assume that f is independent of Z.

Now we consider the LSR for (1) (see [1, Sections 5.3 and 5.4]). We fix an appropriate set of $J_i \in \mathbb{N}$ basis functions $e_i^j \colon \mathbb{R}^{d_1} \to \mathbb{R}$, where $j \in [J_i]$ for $i \in [N]$, and denote $\mathbb{J} := \{J_i\}_{i\in[N]}$. We typically set J_i and e_i^j independent of i, but in general, they may vary for different i. With the time interval $h := T/N$, for $i \in [N-1]$, we define $t_i := ih$, $\Delta W_i := W_{t_{i+1}} - W_{t_i}$,

$$\begin{cases} X_0^N &= x, \\ X_{t_{i+1}}^N &= X_{t_i}^N + b(t_i, X_{t_i}^N)h + \sigma(t_i, X_{t_i}^N)\Delta W_i, \\ Y_T^{N,J} &= g(X_T^N), \\ Y_{t_i}^{N,J} &= \sum_{j=1}^{J_i} \beta_i^j e_i^j(X_{t_i}^N), \\ \{\beta_{i,l}^j\}_j &= \underset{\{\beta^j\}_j \subset \mathbb{R}^{J_i}}{\operatorname{argmin}} \mathbb{E}\left[\left|\sum_{j=1}^{J_i} \beta^j e_i^j(X_{t_i}^N) - \Gamma_l\left(t_i, X_{t_i}^N, Y_{t_{i+1}}^{N,J}\right)\right|^2\right], \end{cases} \tag{2}$$

where $\Gamma_l(t,x,y) = (y_l + hf_l(t,x,y))$. At $t_0 = 0$, $X_0^N = x$, we shall simply define

$$Y_{0,l}^{N,J} := \mathbb{E}\left[\Gamma_l\left(0, x, Y_{t_1,l}^{N,J}\right)\right].$$

Under appropriate assumptions $(X^N, Y^{N,J}, Z^{N,J})$ converges to the solution (X, Y, Z) in L^2. We need to approximate the expectations for (2). In conventional methods, expectations are approximated using Monte Carlo methods. Our approach accelerates this process by using quantum Monte Carlo integration.

3. Quantum algorithm

In this section, we introduce our computational model and present our quantum algorithm for BSDEs, Algorithm 1, which we call the Quantum Least-Square Monte Carlo method (QLSM).

We work in the standard circuit model of quantum computation [2]. We introduce the following definitions and assumptions, which are necessary for calculations in the current problem and similar to those in [3].

Definition 3 (Fixed-point encoding of real numbers) *Let c_1, c_2 be positive integers. For any $(a,b,s) \in \{0,1\}^{c_1} \times \{0,1\}^{c_2} \times \{0,1\} =: \mathcal{I}$, we define*

$$Q(a,b,s) := (-1)^s \Bigg(2^{c_1-1} a_{c_1} + \cdots + 2a_2 + a_1 + \frac{1}{2} b_1 + \cdots + \frac{1}{2^{c_2}} b_{c_2} \Bigg).$$

We denote by $\mathcal{E}$ the set of real numbers written as $Q(a,b,s)$ with some $(a,b,s) \in \mathcal{I}$. For $x = Q(a,b,s) \in \mathcal{E}$, we denote by $|x\rangle$ the computational basis state $|s\rangle \otimes |a_{c_1}\rangle \otimes \cdots \otimes |a_1\rangle \otimes |b_1\rangle \otimes \cdots \otimes |b_{c_2}\rangle$. For $y = (y_1, \ldots, y_n) \in \mathcal{E}^n$, we denote $|y\rangle := |y_1\rangle \otimes \cdots \otimes |y_n\rangle$.

Definition 4 (Quantum arithmetic model) *Given c_1, $c_2 \in \mathbb{N}$ specifying fixed-point precision numbers as in Definition 3, we say that we use a quantum arithmetic model of computation if we have access to quantum circuits for the four arithmetic operations (e.g., a circuit that transforms $|x\rangle|y\rangle$ to $|x\rangle|x+y\rangle$ for x, $y \in \mathcal{E}$ such that $x + y \in \mathcal{E}$) running in $O(1)$ time.*

Definition 5 (Quantum access to a function) *Let $\lambda \colon \mathbb{R}^n \to \mathbb{R}^m$, and let $\lambda_q \colon \mathcal{E}^n \to \mathcal{E}^m$ be a map such that for $x \in \mathcal{E}^n$, $\lambda_q(x)$ is the number in $\mathcal{E}^m$ closest to $\lambda(x)$ in the Euclidean norm. We say that we have $(V_\lambda, \tau_\lambda)$-quantum access to λ if we have access to a quantum circuit V_λ such that for all $x \in \mathcal{E}^m$, $V_\lambda |x\rangle|0\rangle = |x\rangle|\lambda_q(x)\rangle$, and one application of V_λ takes time $m\tau_\lambda$.*

Definition 6 (Quantum access to normal distribution) *Let W_1 be a normally distributed random variable and denote its probability density function by ϕ_{W_1}. Suppose we have a quantum circuit V_{W_1} such that $V_{W_1}|0\rangle = |W_1\rangle := \sum_{j=1}^{N_\mathcal{E}} \sqrt{P_{W_1,j_1}}\, |w_j^\mathcal{E}\rangle$. Here, we define*

$$P_{W_1,j_1} := \begin{cases} \int_{-\infty}^{(w_1^\mathcal{E}+w_2^\mathcal{E})/2} \phi_{W_1}(w)dw & ;\ j_1 = 1, \\ \int_{(w_{j_1}^\mathcal{E}+w_{j_1+1}^\mathcal{E})/2}^{(w_{j_1}^\mathcal{E}+w_{j_1+1}^\mathcal{E})/2} \phi_{W_1}(w)dw & ;\ j_1 = 2, \ldots, N_\mathcal{E}-1, \\ \int_{(w_{j_1-1}^\mathcal{E}+w_{j_1}^\mathcal{E})/2}^{+\infty} \phi_{W_1}(w)dw & ;\ j_1 = N_\mathcal{E}, \end{cases}$$

where $w_1^\mathcal{E} < \cdots < w_{N_\mathcal{E}}^\mathcal{E}$ are arbitrarily chosen $N_\mathcal{E}$ elements of $\mathcal{E}$. We say that we have quantum access to W_1 in this case. For $W = (W_1, \ldots, W_n)$, where W_i are independent copies of W_1, we let $|W\rangle := |W_1\rangle \cdots |W_n\rangle = \sum_{j\in[N_\mathcal{E}]^n} \sqrt{P_{W,j}}\, |w_j^{\mathcal{E}^n}\rangle$, where we use $j := (j_1, \ldots, j_n) \in [N_\mathcal{E}]^n$, $P_{W,j} := \prod_{k=1}^n P_{W_k,j_k}$ and $w_j^{\mathcal{E}^n} := (w_{j_1}^\mathcal{E}, \ldots, w_{j_n}^\mathcal{E})$. We denote by $n\tau_{samp}$ the time one application of V_W takes.

Definition 7 *We say that we have access to quantum controlled rotations if we have a quantum circuit R that for any $x \in [0,1] \cap \mathcal{E}$ operates as*

$$R|x\rangle|0\rangle = |x\rangle\left(\sqrt{1-x}\,|0\rangle + \sqrt{x}\,|1\rangle\right)$$

in constant time.

Assumption 8 (Arithmetic model) *In our work, we perform arithmetic computations on a quantum computer by using a fixed-point representation for real numbers. We assume that we have enough qubits to store these numbers, which are represented by Definition 3. We also assume that we work with sufficient precision such that numerical errors in the fixed-point representation are negligible and will not impact the final output of our algorithm beyond the desired accuracy. Moreover, we assume the following computation models.*

(1) *We use a quantum arithmetic model of computation.*
(2) *Let $\lambda = b$, σ, f, g. We have $(V_\lambda, \tau_\lambda)$-quantum access to λ.*
(3) *For any $i \in [N-1]$ and $j \in [J_i]$, we have $(V_{e_i^j}, \tau_e)$-quantum access to e_i^j with common τ_e.*
(4) *We have quantum access to the normal distribution.*
(5) *We have access to quantum controlled rotations.*

Proposition 9 (Quantum Monte Carlo integration [4, Theorem 2.5]) *Let X be a random variable on a probability space $(\mathcal{E}^n, 2^{\mathcal{E}^n}, \mathbb{P})$ that takes values in $\mathcal{E}$. Assume that we are given a quantum circuit V_X such that $V_X |0\rangle |0\rangle = \sum_{x\in\mathcal{E}^n} \sqrt{\mathbb{P}(x)}\, |x\rangle |X(x)\rangle$. Additionally, assume that we use a quantum arithmetic model of computation and we have access to quantum controlled rotations. Then, for any $\varepsilon > 0$, there exists a quantum algorithm that outputs $\mu \in \mathrm{R}$ such that*

$$|\mu - \mathbb{E}[X]| \le \varepsilon$$

with probability at least $2/3$, querying V_X

$$O\left(\frac{v}{\varepsilon}\log^{\frac{3}{2}}\left(\frac{v}{\varepsilon}\right)\log\left(\frac{v}{\varepsilon}\right)\right)$$

times, where $v^2 := Var(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. The probability of $2/3$ can be improved to $1-\delta$ for any $\delta > 0$ by the Powering Lemma (see [4, Lemma 2.1]) at the cost of an $O(\log(1/\delta))$ multiplicative factor. We call this algorithm $\mathrm{QMCI}(X, \varepsilon, \delta)$.

We now discuss the QLSM. As preparation, we introduce some quantities that will be used in presenting the QLSM.

Definition 10 *We define the following quantities.*

$$\beta := \max_{i,l,j}\left\{\left|\beta^j_{i,l}\right|, 1\right\},$$

$$E := \max_{i,j}\left\{\left\|e^j_i(X^N_{t_i})\right\|_{L^4}, 1\right\},$$

$$F := \max_{i,l}\left\|f_l\left(t_i, X^N_{t_i}, Y^{N,J}_{t_{i+1}}\right)\right\|_{L_4},$$

$$\sigma_{min} := \min_i\{\sigma_{min}(A_i), 1\},$$

$$J := \max_i\{J_i\},$$

$$\Sigma_A := \max_i\sqrt{Var\left(e^{j_1}_i\left(X^N_{t_i}\right)e^{j_2}_i\left(X^N_{t_i}\right)\right)},$$

$$L^2_y := \max_{i,l}\left\|\Gamma_l\left(t_i, X^N_{t_i}, Y^{N,J}_{t_{i+1},l}\right)e_i\left(X^N_{t_i}\right)\right\|_{L^2},$$

$$L^2_{norm} := \max\{\Sigma_A, L^2_y\}.$$

The QLSM is as shown in Algorithm 1.

Remark 11 (1) *We define*

$$a := \frac{4\beta E^3\sqrt{d_2}J^3}{\sigma^2_{min}}, \qquad y := \frac{2E\sqrt{d_2}J}{\sigma_{min}},$$

$$c := 1 + LT + \frac{LT}{N},$$

$$\delta_A := \frac{\delta}{NJ^2}, \qquad \delta_y := \frac{\delta}{d_2NJ},$$

$$\widetilde{\varepsilon}_A := \frac{\varepsilon}{2e^caN}, \qquad \widetilde{\varepsilon}_y := \frac{\varepsilon}{2e^cyN}.$$

Algorithm 1 Quantum Least-Square Monte Carlo method for BSDEs (f independents on Z)

Input: dimensions $d_1, d_2, d \in \mathbb{N}$, parameters $N, J_i \in \mathbb{N}$, $\varepsilon, \delta \in (0,1)$, initial condition x, quantum access to $\{e^j_i\}$, b, σ, g, f.

1: Define $\widetilde{\varepsilon}'_A$, $\widetilde{\varepsilon}'_y$, δ_A, δ_y with Remark 11.

2: **for** $i = 0$ to $N-1$ **do**

3: Construct the oracle $V_{X_{t_i}}$ that acts as

$$V_{X_{t_i}}|x\rangle|0\rangle|0\rangle = |x\rangle \sum_{K\in[N_{\mathcal{E}}]^{d_1}} \sqrt{P_{\Delta W_i,K}}\left|w^{\mathcal{E}^{d_1}}_K\right\rangle \left|x + b(t_i,x)h + \sigma(t_i,x)w^{\mathcal{E}^{d_1}}_K\right\rangle \quad (3)$$

for any $x \in \mathbb{R}^{d_1}$.

4: For each $j_1, j_2 \in [J_i]$, run $\mathrm{QMCI}\left(e^{j_1}_i\left(X^N_{t_i}\right)e^{j_2}_i\left(X^N_{t_i}\right), \widetilde{\varepsilon}_A, \delta_A\right)$ and let the output be $(\widetilde{A}_i)_{j_1,j_2}$.

5: Compute the inverses $\widetilde{A}^{-1}_i$ by a classical computer.

6: **end for**

7: Define $\widetilde{\mathcal{Y}}^{N,J}_{t_N} := g$.

8: **for** $i = N-1$ to 1 **do**

9: For each $j \in [J_i]$, $l \in [d_2]$, run QMCI $\left(\Gamma_l\left(t_i, X^N_{t_i}, \widetilde{\mathcal{Y}}^{N,J}_{t_{i+1},l}\right)e^j_i\left(X^N_{t_i}\right), \widetilde{\varepsilon}_y, \delta_y\right)$ and let the output be $\widetilde{y}^j_{i,l}$.

10: Compute $\widetilde{\beta}_{i,l} = \widetilde{A}^{-1}_i\widetilde{y}_{i,l}$ for each $l \in [d_2]$ by a classical computer and define $\widetilde{\mathcal{Y}}^{N,J}_{t_i} := \sum_{j=1}^{J_i}\widetilde{\beta}^j_i e^j_i$.

11: **end for**

12: For each $l \in [d_2]$, run $\mathrm{QMCI}\left(\Gamma_l\left(0, x_0, \widetilde{\mathcal{Y}}^{N,J}_{t_1,l}\right), \widetilde{\varepsilon}_y, \delta_y\right)$ and let the output be $\widetilde{Y}^{N,J}_{0,l}$.

Output: $\widetilde{Y}^{N,J}_0$.

(2) *$V_{X_{t_i}}$ uses V_b, V_σ, and V_{W_i} once each. Thus, the time τ_X for one query to it is $O(d_1\tau_b + d_1d\tau_\sigma + d\tau_{samp})$.*

(3) *We will now explain the oracles used in the QMCIs in Algorithm 1 in more detail. First, let us consider step 4. We construct the oracle to generate the state*

$$|x\rangle \sum_{K_0,\ldots,K_{i-1}\in[N_{\mathcal{E}}]^{d_1}} \sqrt{P_{\Delta W_0,K_0}\cdots P_{\Delta W_{i-1},K_{i-1}}} \left|w^{\mathcal{E}^{d_1}}_{K_0}\right\rangle\cdots\left|w^{\mathcal{E}^{d_1}}_{K_{i-1}}\right\rangle\left|X^N_{t_i}\left(w^{\mathcal{E}^{d_1}}_{K_0},\ldots,w^{\mathcal{E}^{d_1}}_{K_{i-1}}\right)\right\rangle,$$

where we have omitted some registers for simplicity, and $X^N_{t_i}\left(w^{\mathcal{E}^{d_1}}_{K_0},\ldots,w^{\mathcal{E}^{d_1}}_{K_{i-1}}\right)$ is generated by (3) with $\Delta W_0 = w^{\mathcal{E}^{d_1}}_{K_0},\ldots,\Delta W_{i-1} = w^{\mathcal{E}^{d_1}}_{K_{i-1}}$. Combining this with $V_{e^{j_1}_i}$ and $V_{e^{j_2}_i}$, we can construct the oracle $V_{(\widetilde{A}_i)_{j_1,j_2}}$ to generate

$$|x\rangle \sum_{K_0,\ldots,K_{i-1}\in[N_{\mathcal{E}}]^{d_1}} \sqrt{P_{\Delta W_0,K_0}\cdots P_{\Delta W_{i-1},K_{i-1}}} \left|w^{\mathcal{E}^{d_1}}_{K_0}\right\rangle\cdots\left|w^{\mathcal{E}^{d_1}}_{K_{i-1}}\right\rangle\left|X^N_{t_i}\right\rangle\left|e^{j_1}_i(X^N_{t_i})e^{j_2}_i(X^N_{t_i})\right\rangle,$$

where we omit the arguments of $X^N_{t_i}$ for simplicity. Equipped with $V_{(\widetilde{A}_i)_{j_1,j_2}}$, we can run $QMCI(e_i^{j_1}(X^N_{t_i})e_i^{j_2}(X^N_{t_i}),\varepsilon_A,\delta_A)$. In $V_{(\widetilde{A}_i)_{j_1,j_2}}$ we query $V_{X_0}\cdots V_{X_{i-1}}, V_{e_i^{j_1}}$, and $V_{e_i^{j_2}}$ once each, which takes $O(N\tau_X+\tau_e)$ time.
We can consider other QMCIs similarly.
$QMCI(\Gamma_l(t_i,X^N_{t_i},\widetilde{\mathcal{Y}}^{N,J}_{t_{i+1},l})e_i^j(X^N_{t_i}),\varepsilon_y,\delta_y)$ in step 9 is enabled by the oracle to generate

$$|x\rangle \sum_{K_0,\ldots,K_i\in[N_{\mathcal{E}}]^{d_1}} \sqrt{P_{\Delta W_0,K_0}\cdots P_{\Delta W_i,K_i}} \left|w^{\mathcal{E}^{d_1}}_{K_0}\right\rangle\cdots\left|w^{\mathcal{E}^{d_1}}_{K_i}\right\rangle \left|\Gamma_l\left(t_i,X^N_{t_i},\widetilde{\mathcal{Y}}^{N,J}_{t_{i+1},l}\right)e_i^j\left(X^N_{t_i}\right)\right\rangle,$$

which we implement by using $V_{X_0}\cdots V_{X_{t_i}}$, and V_{f_l} once each and oracles in $\{V_{e_i^j}\}$ $O(J)$ times. One query to this oracle takes $O(N\tau_X+J\tau_e+\tau_f)$ time. The QMCIs in step 12 are similar to those in step 9.

(4) *We need to consider g only in the first round in the loop from steps 8 to 11, so we regard the cost of quantum access to g as constant throughout Algorithm 1.*

(5) *We require an L^4-solution (X,Y,Z) to ensure $L^2_{norm}<\infty$.*

4. Error analysis and cost evaluation

In our algorithm, error estimation can be divided into two parts,

$$\left|Y_0-\widetilde{Y}^{N,J}_0\right| \le \left|Y_0-Y^{N,J}_0\right| + \left|Y^{N,J}_0-\widetilde{Y}^{N,J}_0\right|.$$

The first part has already been estimated in previous work [1], so we estimate only our main target $|Y^{N,J}_0-\widetilde{Y}^{N,J}_0|$.

Considering QMCI, we introduce the following temporary assumptions.

Assumption 12 *We can get $\widetilde{A}_i=(\widetilde{A}_i^{j_1,j_2})_{j_1,j_2}$ and $\widetilde{y}_{i,l}=(\widetilde{y}^j_{i,l})_j$ such that for some ε_A, $\varepsilon_y>0$,*

$$\left|\mathbb{E}\left[e_i^{j_1}\left(X^N_{t_i}\right)e_i^{j_2}\left(X^N_{t_i}\right)\right]-\widetilde{A}_i^{j_1,j_2}\right|\le\varepsilon_A,$$
$$\left|\mathbb{E}\left[\Gamma_l\left(t_i,X^N_{t_i},\widetilde{Y}^{N,J}_{t_{i+1},l}\right)e_i^j\left(X^N_{t_i}\right)\right]-\widetilde{y}^j_{i,l}\right|\le\varepsilon_y$$

hold for each $i\in[N-1]$, $l\in[d_2]$, where

$$\widetilde{Y}^{N,J}_{t_i}:=\sum_{j=1}^{J_i}\widetilde{\beta}_i^j e_i^j\left(X^N_{t_i}\right),\qquad \widetilde{\beta}_i:=\widetilde{A}_i^{-1}\widetilde{Y}_{t_i}.$$

Additionally, we can get $\widetilde{Y}^{N,J}_{0,l}$ satisfying

$$\left|\mathbb{E}\left[\Gamma_l\left(0,x,\widetilde{Y}^{N,J}_{t_1,l}\right)\right]-\widetilde{Y}^{N,J}_{0,l}\right|\le\varepsilon_y.$$

Finally, we present our two main theorems.

Theorem 13 (Global error estimate) *Let Assumptions 1 and 12 hold and fix $p=4$. We assume that $\varepsilon_A\le\sigma^2_{min}/2E\sqrt{d_2}\,NJ^2$. Then, for $h\le 1/F$ the equation*

$$\max_{i\in[N-1]}\left\|Y^{N,J}_{t_i}-\widetilde{Y}^{N,J}_{t_i}\right\|_{L^2}\le Ne^c(a\varepsilon_A+y\varepsilon_y)$$

holds.

Here, we choose $\widetilde{\varepsilon}_A$, $\widetilde{\varepsilon}_y$, δ_A, and δ_y according to Remark 11, and we get (4) by using Theorem 13.

Theorem 14 *Let Assumptions 1 and 8 hold and fix $p=4$. Then, for all ε, $\delta>0$ and for $h\le\min\{1/F,1/L\}$, Algorithm 1 outputs $\{\widetilde{Y}^{N,J}_{i,l}\}_{i\in[N-1]_+0,l\in[d_2]}$ satisfying*

$$\max_{i\in[N-1]_+0}\left\|Y^{N,J}_{t_i}-\widetilde{Y}^{N,J}_{t_i}\right\|_{L^2}\le\varepsilon \tag{4}$$

with probability at least $1-\delta$ in time

$$O\Bigg(\frac{e^c(L^2+\eta_y)\beta E^3 d_2\sqrt{d_2}\,N^3J^5}{\sigma^2_{min}}\frac{1}{\varepsilon}\log\left(\frac{d_2NJ^2}{\delta}\right) \times\log^{\frac{3}{2}}\left(\frac{e^c(L^2+\eta_y)\beta E^3\sqrt{d_2}\,NJ^3}{\sigma^2_{min}}\right) \times\log\left(\frac{e^c(L^2+\eta_y)\beta E^3\sqrt{d_2}\,NJ^3}{\sigma^2_{min}}\right)(\tau_X+\tau_e+\tau_f)\Bigg),$$

where $\eta_y:=20\sqrt{d_2}\sqrt{J}\,JE^2/\sigma_{min}\varepsilon$ and L^2_{norm}, c are defined in Definition 10 and Remark 11, respectively.

Remark 15 *Theorem 13 applies not only to quantum algorithms but also to all algorithms that satisfy Assumption 12, that is, algorithms that approximate expected values. Therefore, by using Chebyshev's inequality, the cost of the classical Monte Carlo method can also be estimated in a similar fashion. As a result, QLSM achieves a speedup compared with the estimation given in [5].*

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